,VISVESVARAYA TECHNOLOGICAL UNIVERSITY

"JnanaSangama", Belgaum -590014, Karnataka.

LAB REPORT
on

Machine Learning (23CS6PCMAL)

Submitted by

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in partial fulfillment for the award of the degree of

BACHELOR OF ENGINEERING
in
COMPUTER SCIENCE AND ENGINEERING

B.M.S. COLLEGE OF ENGINEERING
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B.M.S. College of Engineering,
Bull Temple Road, Bangalore 560019
(Affiliated To Visvesvaraya Technological University, Belgaum)
Department of Computer Science and Engineering

CERTIFICATE

This is to certify that the Lab work entitled "Machine Learning (23CS6PCMAL)" carried out by **Rayhan Sadat (1BM23CS264),** who is bonafide student of **B.M.S. College of Engineering.** It is in partial fulfillment for the award of **Bachelor of Engineering in Computer Science and Engineering** of the Visvesvaraya Technological University, Belgaum. The Lab report has been approved as it satisfies the academic requirements in respect of an Machine Learning (23CS6PCMAL) work prescribed for the said degree.

Lab faculty Incharge Name Assistant Professor Department of CSE, BMSCE	Rashmi H Assistant Professor Department of CSE, BMSCE

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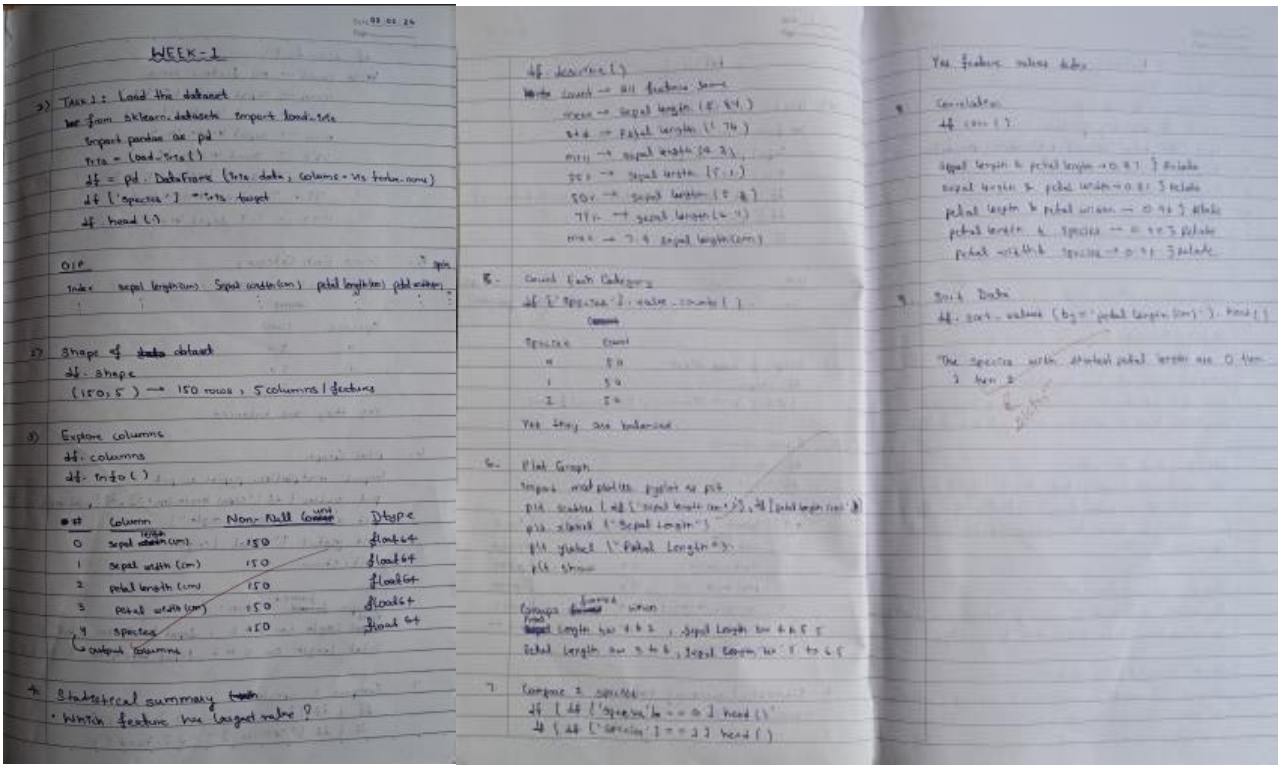
Github Link:
https://github.com/RayhanSadat26/ML_Lab

Program 1

Screenshot



Code:

In [12]:
```
from sklearn.datasets import load_iris
import pandas as pd
import matplotlib.pyplot as plt
```

In [6]:
```
iris = load_iris()
df =pd.DataFrame(iris.data,columns=iris["feature_names"])
df["species"] = iris.target
df.head()
```

Out[6]:

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	species
0	5.1	3.5	1.4	0.2	0
1	4.9	3.0	1.4	0.2	0
2	4.7	3.2	1.3	0.2	0
3	4.6	3.1	1.5	0.2	0
4	5.0	3.6	1.4	0.2	0

In [7]:
```
df.shape
```

Out[7]:
```
(150, 5)
```

In [8]:
```
df.columns
```

Out[8]:
```
Index(['sepal length (cm)', 'sepal width (cm)', 'petal length (cm)',
       'petal width (cm)', 'species'],
      dtype='object')
```

In [9]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
 #   Column             Non-Null Count  Dtype  
---  ------             --------------  -----  
 0   sepal length (cm)  150 non-null    float64
 1   sepal width (cm)   150 non-null    float64
 2   petal length (cm)  150 non-null    float64
 3   petal width (cm)   150 non-null    float64
 4   species            150 non-null    int64  
dtypes: float64(4), int64(1)
memory usage: 6.0 KB
```

In [10]:
```
df.describe()
```

Out[10]:

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	species
count	150.000000	150.000000	150.000000	150.000000	150.000000
mean	5.843333	3.057333	3.758000	1.199333	1.000000
std	0.828066	0.435866	1.765298	0.762238	0.819232
min	4.300000	2.000000	1.000000	0.100000	0.000000
25%	5.100000	2.800000	1.600000	0.300000	0.000000
50%	5.800000	3.000000	4.350000	1.300000	1.000000
75%	6.400000	3.300000	5.100000	1.800000	2.000000
max	7.900000	4.400000	6.900000	2.500000	2.000000

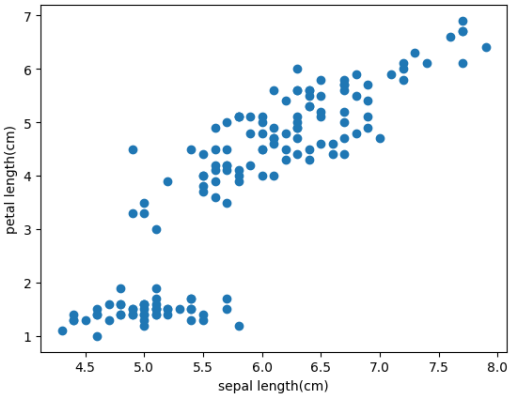
In [11]:
```
df['species'].value_counts()
```

Out[11]:

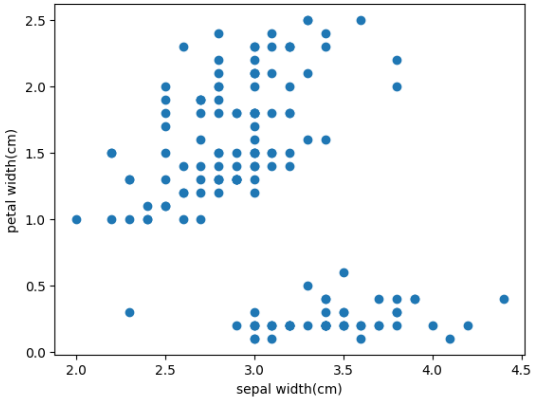
species	count
0	50
1	50
2	50

dtype: int64

In [13]:
```
plt.scatter(df['sepal length (cm)'],df['petal length (cm)'])
plt.xlabel('sepal length(cm)')
plt.ylabel('petal length(cm)')
plt.show()
```



In [15]:
```
plt.scatter(df['sepal width (cm)'],df['petal width (cm)'])
plt.xlabel('sepal width(cm)')
plt.ylabel('petal width(cm)')
plt.show()
```



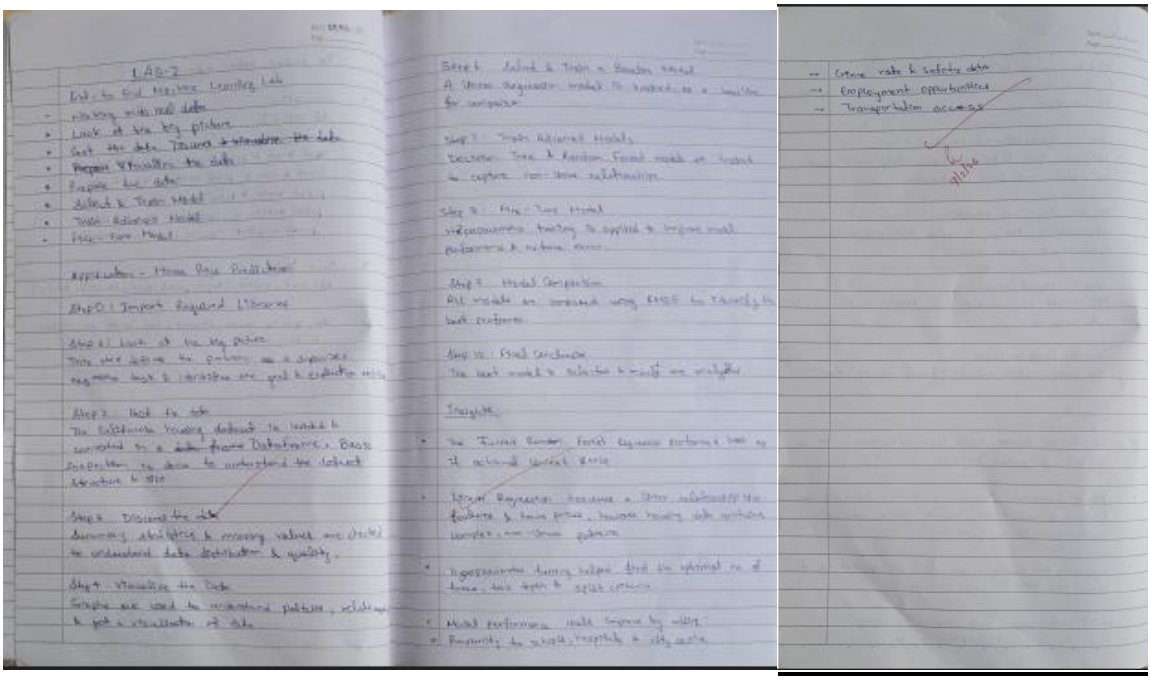
In [16]:
```
df[df.species==0].head()
```

Out[16]:

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	species
0	5.1	3.5	1.4	0.2	0
1	4.9	3.0	1.4	0.2	0
2	4.7	3.2	1.3	0.2	0
3	4.6	3.1	1.5	0.2	0
4	5.0	3.6	1.4	0.2	0

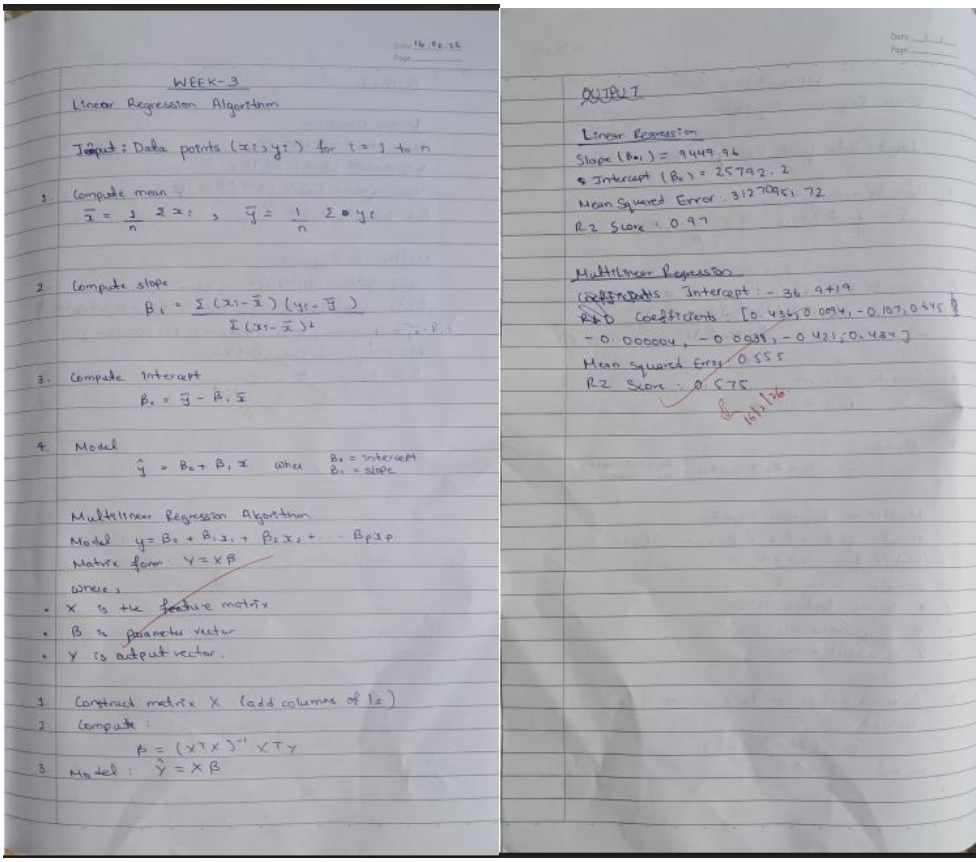
Program 2

Screenshots:



Program 3 & 4

Screenshot



WEEK-3

Linear Regression Algorithm

Input : Data points (x_i, y_i) for $i = 1$ to n

1. Compute mean

$$\bar{x} = \frac{1}{n} \sum x_i \quad , \quad \bar{y} = \frac{1}{n} \sum y_i$$

2. Compute slope

$$B_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$$

3. Compute Intercept

$$B_0 = \bar{y} - B_1 \bar{x}$$

4. Model

$$\hat{y} = B_0 + B_1 x \quad \text{where} \quad B_0 = \text{intercept}, \; B_1 = \text{slope}$$

Multilinear Regression Algorithm

Model $y = B_0 + B_1 x_1 + B_2 x_2 + \cdots B_p x_p$

Matrix form $Y = X\beta$

where,

- X is the feature matrix
- β is parameter vector
- Y is output vector.

1. Construct matrix X (add columns of 1s)
2. Compute :

$$\beta = (X^T X)^{-1} X^T y$$

3. Model : $\hat{Y} = X\beta$

OUTPUT

Linear Regression

Slope (B_1) = 9449.96

Intercept (B_0) = 25792.2

Mean Squared Error : 31270951.72

R2 Score : 0.47

Multilinear Regression

Coefficients : Intercept : -36.9419

Coefficients : [0.4365, 0.0094, -0.107, 0.645, -0.000004, -0.0039, -0.421, -0.434]

Mean Squared Error : 0.555

R2 Score : 0.575

Code:

```
from sklearn.datasets import fetch_california_housing
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split

# Load dataset
data = fetch_california_housing()
X = data.data[:, [0]]   # Use only MedInc (first feature)
y = data.target

# Split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Train
model = LinearRegression()
model.fit(X_train, y_train)

print("Intercept:", model.intercept_)
print("Slope:", model.coef_[0])
```

```
print("R^2:", model.score(X_test, y_test))
```

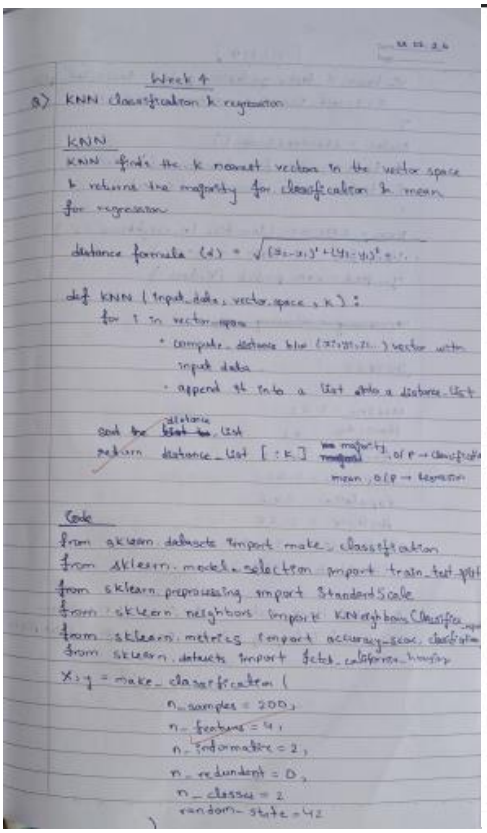
O/P:
Intercept: 0.4445972916907872
Slope: 0.4193384939381273
R^2: 0.45885918903846656

```
from sklearn.datasets import fetch_california_housing
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split

# Load dataset
data = fetch_california_housing()
X = data.data   # All features
y = data.target

# Split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Train
model = LinearRegression()
model.fit(X_train, y_train)

print("Intercept:", model.intercept_)
print("Coefficients:", model.coef_)
print("R^2:", model.score(X_test, y_test))
```

O/P:
Intercept: -37.023277706063794
Coefficients: [4.48674910e-01 9.72425752e-03 -1.23323343e-01 7.83144907e-01
 -2.02962058e-06 -3.52631849e-03 -4.19792487e-01 -4.33708065e-01]
R^2: 0.5757877060324524

Program 5

Screenshot


Week 4

8) KNN Classification & regression

KNN

KNN finds the k nearest vectors in the vector space & returns the majority for classification & mean for regression

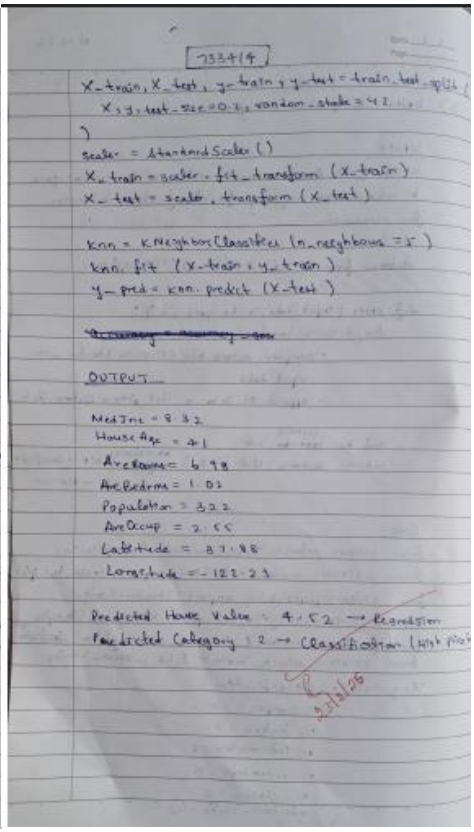
distance formula (d) = $\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$

def KNN (input_data, vector_space, k):
for i in vector_space:
- compute distance b/w (x1,y1,z1,...) vector with input data
- append it into a list called a distance list

sort the distance list
return distance_list [:k] the majority of it → classification
mean of it → regression

Code

```
from sklearn.datasets import make_classification
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScale
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score, classification
from sklearn.datasets import fetch_california_housing
X, y = make_classification(
    n_samples = 200,
    n_features = 4,
    n_informative = 2,
    n_redundant = 0,
    n_classes = 2
    random_state = 42
)
```


```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size = 0.2, random_state = 42
)
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

knn = KNeighborsClassifier(n_neighbors = 5)
knn.fit(X_train, y_train)
y_pred = knn.predict(X_test)
```

OUTPUT

```
MedInc = 8.32
HouseAge = 41
AveRooms = 6.98
AveBedrms = 1.02
Population = 322
AveOccup = 2.55
Latitude = 37.88
Longitude = -122.23

Predicted House Value : 4.52 → Regression
Predicted Category : 2 → Classification
```

Code:

```
import numpy as np

from sklearn.neighbors import KNeighborsClassifier, KNeighborsRegressor

X = np.array([[100], [105], [98], [150], [160], [155]])

y_class = np.array([0, 0, 0, 1, 1, 1])

y_reg = np.array([100, 105, 98, 150, 160, 155])

# 1) KNN Classification

knn_classifier = KNeighborsClassifier(n_neighbors=3)

knn_classifier.fit(X, y_class)

new_ball = np.array([[110]])
```

```
class_prediction = knn_classifier.predict(new_ball)
if class_prediction[0] == 0:
    print("Classification: Red Ball")
else:
    print("Classification: Blue Ball")
# 2) KNN Regression
knn_regressor = KNeighborsRegressor(n_neighbors=3)
knn_regressor.fit(X, y_reg)
weight_prediction = knn_regressor.predict(new_ball)
print("Regression (Average Predicted Weight):", weight_prediction[0])
```

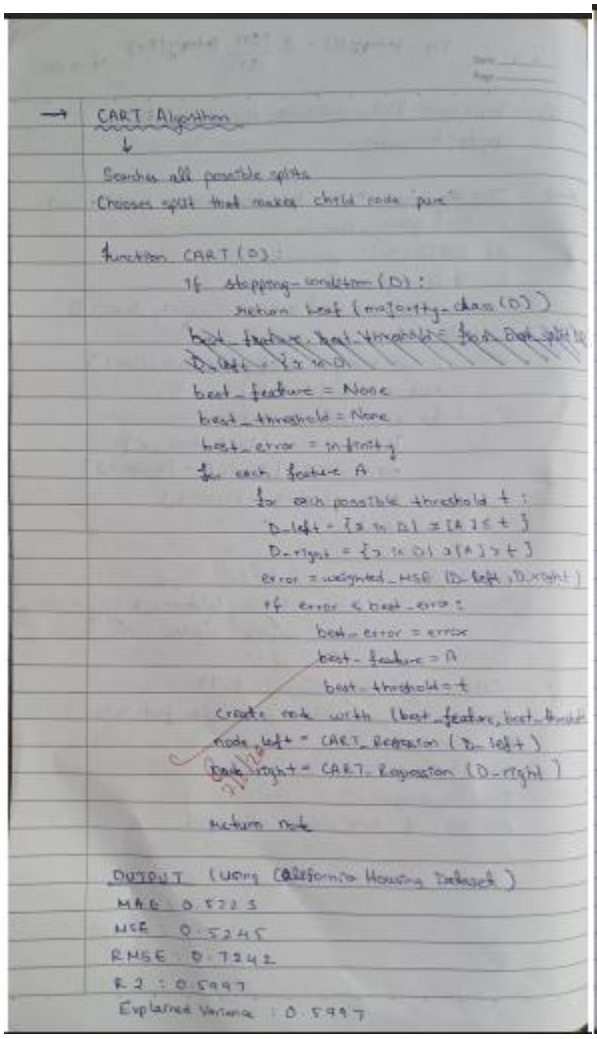
O/P:

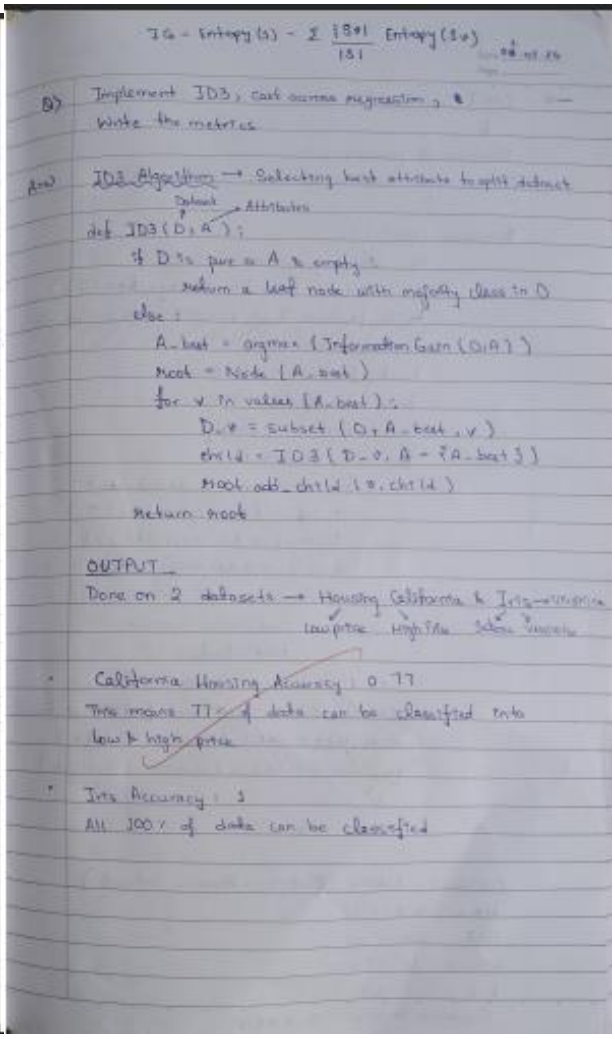
Classification: Red Ball

Regression (Average Predicted Weight): 101.0

Program 6

Screenshots:





Code:

```
import numpy as np

import pandas as pd

from sklearn.datasets import load_iris, load_breast_cancer

from sklearn.model_selection import train_test_split

from sklearn.tree import DecisionTreeClassifier

from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score, confusion_matrix
```

```
def evaluate_classification(X, y, dataset_name):
    print("\n==================================")
    print("Dataset:", dataset_name)
    print("==================================")
# Split dataset
    X_train, X_test, y_train, y_test = train_test_split(
        X, y, test_size=0.3, random_state=42
    )
 # ID3 → entropy criterion
    model = DecisionTreeClassifier(criterion="entropy", random_state=42)
    model.fit(X_train, y_train)
    y_pred = model.predict(X_test)
 # Error Metrics
    accuracy = accuracy_score(y_test, y_pred)
    precision = precision_score(y_test, y_pred, average='weighted')
    recall = recall_score(y_test, y_pred, average='weighted')
    f1 = f1_score(y_test, y_pred, average='weighted')
    cm = confusion_matrix(y_test, y_pred)
    print("Accuracy:", accuracy)
    print("Precision:", precision)
    print("Recall:", recall)
    print("F1 Score:", f1)
    print("\nConfusion Matrix:")
    print(cm)
```

 # Dataset 1: Iris

iris = load_iris()

evaluate_classification(iris.data, iris.target, "Iris Dataset")

 # Dataset 2: Breast Cancer

cancer = load_breast_cancer()

evaluate_classification(cancer.data, cancer.target, "Breast Cancer Dataset")

O/P:

==========

Dataset: Iris Dataset

==========

Accuracy: 0.9777777777777777

Precision: 0.9793650793650793

Recall: 0.9777777777777777

F1 Score: 0.9777448559670783

Confusion Matrix:

[[19 0 0]

[0 13 0]

[0 1 12]]

==========

Dataset: Breast Cancer Dataset

==========

Accuracy: 0.9649122807017544

Precision: 0.9649541140481607

Recall: 0.9649122807017544

F1 Score: 0.9647902680643604

Confusion Matrix:

[[59 4]

 [2 106]]

```
# Decision Tree Regression using CART Algorithm
import numpy as np
from sklearn.datasets import fetch_california_housing, load_diabetes
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeRegressor
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
def evaluate_regression(X, y, dataset_name):
    print("\n====================================")
    print("Dataset:", dataset_name)
    print("====================================")
    X_train, X_test, y_train, y_test = train_test_split(
        X, y, test_size=0.3, random_state=42
    )
# CART → squared_error criterion
    model = DecisionTreeRegressor(
        criterion="squared_error",
```

```
        random_state=42
    )
    model.fit(X_train, y_train)
    y_pred = model.predict(X_test)
 # Error Metrics
    mae = mean_absolute_error(y_test, y_pred)
    mse = mean_squared_error(y_test, y_pred)
    rmse = np.sqrt(mse)
    r2 = r2_score(y_test, y_pred)
    print("MAE:", mae)
    print("MSE:", mse)
    print("RMSE:", rmse)
    print("R2 Score:", r2)
# Dataset 1: California Housing
housing = fetch_california_housing()
evaluate_regression(housing.data, housing.target, "California Housing Dataset")
# Dataset 2: Diabetes
diabetes = load_diabetes()
evaluate_regression(diabetes.data, diabetes.target, "Diabetes Dataset")
```

O/P:

```
==================================
Dataset: California Housing Dataset
==================================
MAE: 0.46904822189922485
```

MSE: 0.5280096503174904

RMSE: 0.7266427253592307

R2 Score: 0.5977192261218356

==========

Dataset: Diabetes Dataset

==========

MAE: 60.015037593984964

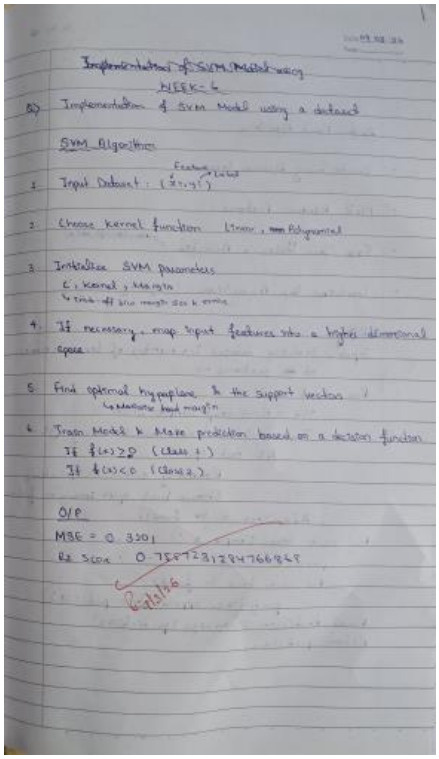
MSE: 5697.789473684211

RMSE: 75.48370336492647

R2 Score: -0.05547696148647652

Program 7

Screenshots:



Code:

```
import numpy as np

import pandas as pd

from sklearn.datasets import load_iris

from sklearn.model_selection import train_test_split

from sklearn.preprocessing import StandardScaler

from sklearn.svm import SVC

from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

# Load dataset

iris = load_iris()
```

```
X = iris.data
y = iris.target
# Train-test split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)
# Feature scaling
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
# Create SVM model
svm_model = SVC(kernel='linear')
# Train model
svm_model.fit(X_train, y_train)
# Predictions
y_pred = svm_model.predict(X_test)
# Evaluation
print("Accuracy:", accuracy_score(y_test, y_pred))
print("\nConfusion Matrix")
print(confusion_matrix(y_test, y_pred))
print("\nClassification Report")
print(classification_report(y_test, y_pred))
```

O/P:

Accuracy: 0.9666666666666667

Confusion Matrix

[[10 0 0]

[0 8 1]

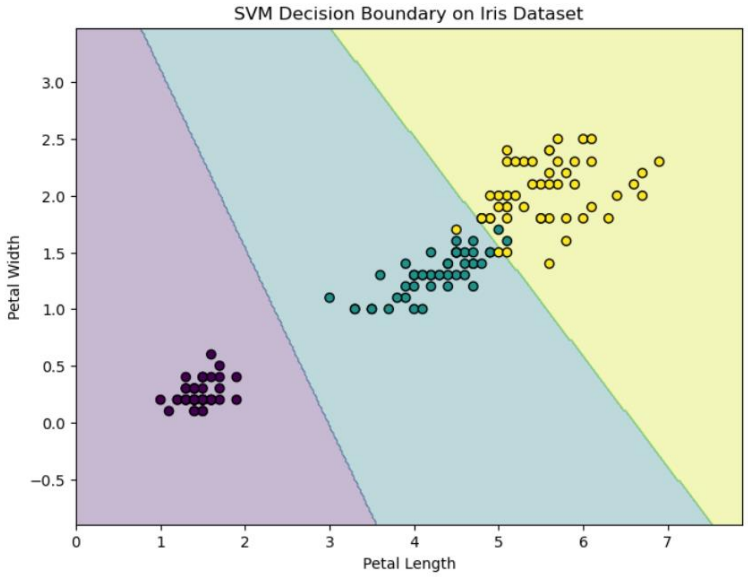
[0 0 11]]

Classification Report

	precision	recall	f1-score	support
0	1.00	1.00	1.00	10
1	1.00	0.89	0.94	9
2	0.92	1.00	0.96	11
accuracy			0.97	30
macro avg	0.97	0.96	0.97	30
weighted avg	0.97	0.97	0.97	30

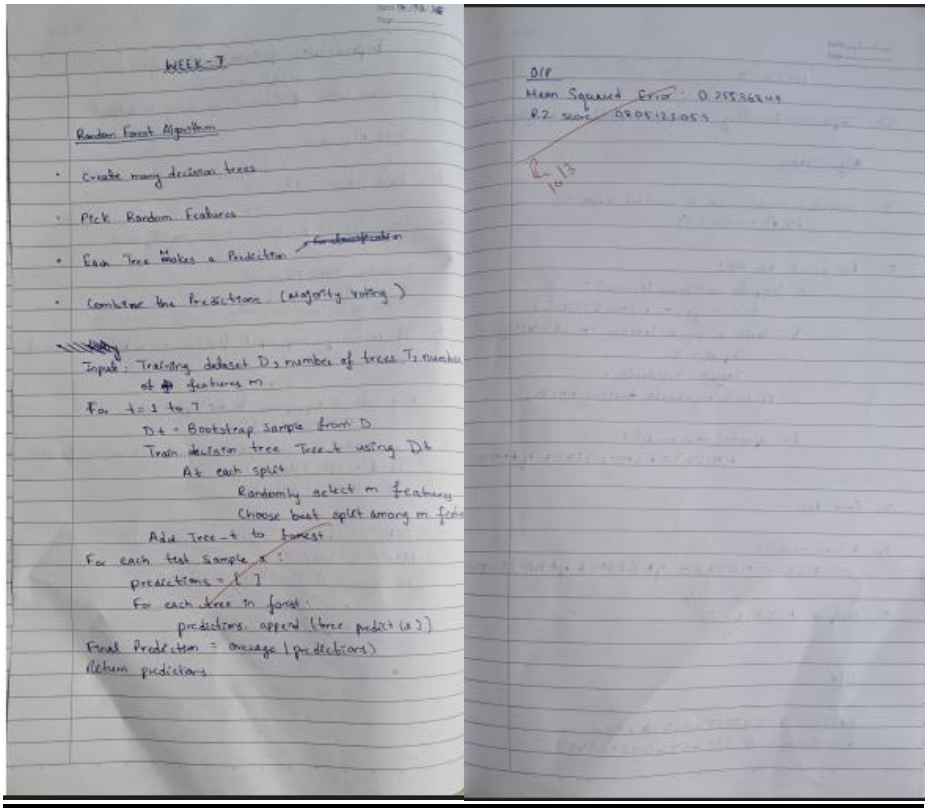
```
import numpy as np
import matplotlib.pyplot as plt
from sklearn import datasets
from sklearn.svm import SVC
iris = datasets.load_iris()
X = iris.data[:, 2:4]   # petal length and petal width
y = iris.target
model = SVC(kernel='linear')
model.fit(X, y)
SVC(kernel='linear')
```

```
x_min, x_max = X[:,0].min() - 1, X[:,0].max() + 1
y_min, y_max = X[:,1].min() - 1, X[:,1].max() + 1
xx, yy = np.meshgrid(
    np.arange(x_min, x_max, 0.02),
    np.arange(y_min, y_max, 0.02)
)
Z = model.predict(np.c_[xx.ravel(), yy.ravel()])
Z = Z.reshape(xx.shape)
plt.figure(figsize=(8,6))
plt.contourf(xx, yy, Z, alpha=0.3)
scatter = plt.scatter(X[:,0], X[:,1], c=y, edgecolors='k')
plt.xlabel("Petal Length")
plt.ylabel("Petal Width")
plt.title("SVM Decision Boundary on Iris Dataset")
plt.show()
```



Program 8

Screenshots



Code:

```
import pandas as pd

from sklearn.model_selection import train_test_split

from sklearn.ensemble import RandomForestClassifier

from sklearn.metrics import accuracy_score, classification_report

import warnings

warnings.filterwarnings('ignore')

# Load dataset

titanic_data = pd.read_csv('titanic.csv')

# Remove rows where Survived is missing
```

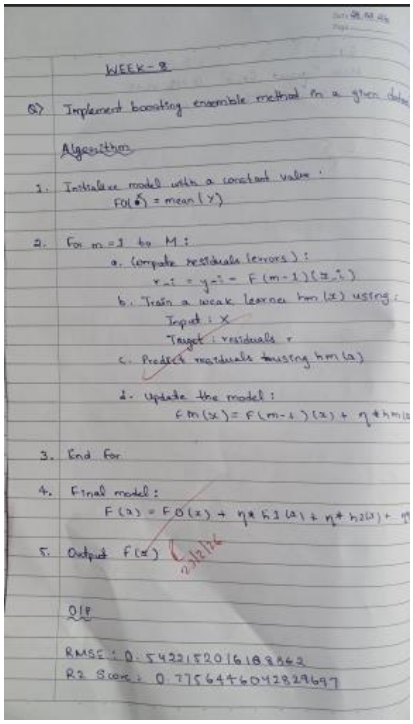
```
titanic_data = titanic_data.dropna(subset=['Survived'])
# Select features and target
X = titanic_data[['Pclass', 'Sex', 'Age', 'SibSp', 'Parch', 'Fare']].copy()
y = titanic_data['Survived']
# Encode categorical variable
X['Sex'] = X['Sex'].map({'female': 0, 'male': 1})
# Fill missing Age values with median
X['Age'] = X['Age'].fillna(X['Age'].median())
# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
# Create Random Forest model
rf_classifier = RandomForestClassifier(
    n_estimators=100,
    random_state=42
)
# Train model
rf_classifier.fit(X_train, y_train)
# Make predictions
y_pred = rf_classifier.predict(X_test)
# Evaluate model
accuracy = accuracy_score(y_test, y_pred)
classification_rep = classification_report(y_test, y_pred)
```

```
print(f"Accuracy: {accuracy:.2f}")
print("\nClassification Report:\n", classification_rep)
# Test with a sample passenger
sample = X_test.iloc[[0]]
prediction = rf_classifier.predict(sample)
sample_dict = sample.iloc[0].to_dict()
print("\nSample Passenger:", sample_dict)
print("Predicted Survival:", "Survived" if prediction[0] == 1 else "Did Not Survive")
```

```
Accuracy: 0.80

Classification Report:
              precision    recall  f1-score   support

           0       0.82      0.85      0.83       105
           1       0.77      0.73      0.75        74

    accuracy                           0.80       179
   macro avg       0.79      0.79      0.79       179
weighted avg       0.80      0.80      0.80       179

Sample Passenger: {'Pclass': 3.0, 'Sex': 1.0, 'Age': 28.0, 'SibSp': 1.0, 'Parch': 1.0, 'Fare': 15.2458}
Predicted Survival: Did Not Survive
```

Program 9

Screenshots



Code:

 # Step 1: Import libraries

import numpy **as** np

import pandas **as** pd

import matplotlib.pyplot **as** plt

from sklearn.datasets **import** load_breast_cancer

from sklearn.model_selection **import** train_test_split

from sklearn.ensemble **import** AdaBoostClassifier

from sklearn.tree **import** DecisionTreeClassifier

from sklearn.decomposition **import** PCA

from sklearn.metrics **import** accuracy_score, classification_report, confusion_matrix

```
# Step 2: Load dataset
data = load_breast_cancer()
X = data.data
y = data.target
# Step 3: Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
# Step 4: Create base learner
base_model = DecisionTreeClassifier(max_depth=1)
# Step 5: Create AdaBoost model
model = AdaBoostClassifier(
    estimator=base_model,
    n_estimators=50,
    learning_rate=1.0,
    random_state=42
)
# Step 6: Train model
model.fit(X_train, y_train)
# Step 7: Predictions
y_pred = model.predict(X_test)
# Step 8: Evaluation
print("Accuracy:", accuracy_score(y_test, y_pred))
print("\nConfusion Matrix:\n", confusion_matrix(y_test, y_pred))
```

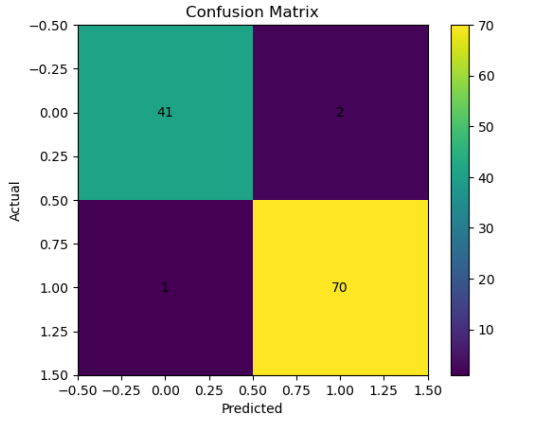
```
print("\nClassification Report:\n", classification_report(y_test, y_pred))

# ------------------------------

# 📊 VISUALIZATION 1: Confusion Matrix Heatmap

# ------------------------------

cm = confusion_matrix(y_test, y_pred)

plt.figure()

plt.imshow(cm)

plt.title("Confusion Matrix")

plt.xlabel("Predicted")

plt.ylabel("Actual")

for i in range(len(cm)):

    for j in range(len(cm)):

        plt.text(j, i, cm[i, j], ha='center', va='center')

plt.colorbar()

plt.show()

# ------------------------------

# 📊 VISUALIZATION 2: Error Reduction Plot

# ------------------------------

errors = []

for y_pred_stage in model.staged_predict(X_test):

    errors.append(np.mean(y_pred_stage != y_test))

plt.figure()

plt.plot(errors)

plt.title("Error Reduction over Boosting Iterations")
```

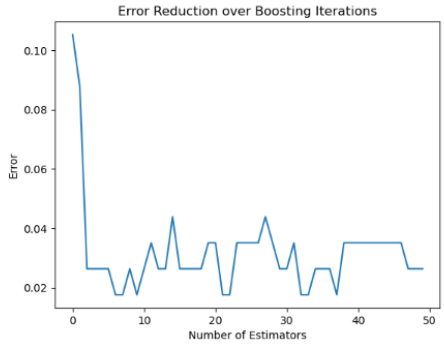
```
plt.xlabel("Number of Estimators")
plt.ylabel("Error")
plt.show()
# ------------------------------
# 🔷 VISUALIZATION 3: PCA Decision Boundary
# ------------------------------
# Reduce to 2D
pca = PCA(n_components=2)
X_pca = pca.fit_transform(X)
# Train again on PCA data
model_pca = AdaBoostClassifier(
    estimator=DecisionTreeClassifier(max_depth=1),
    n_estimators=50,
    random_state=42
)
model_pca.fit(X_pca, y)
# Plot boundary
x_min, x_max = X_pca[:, 0].min() - 1, X_pca[:, 0].max() + 1
y_min, y_max = X_pca[:, 1].min() - 1, X_pca[:, 1].max() + 1
xx, yy = np.meshgrid(
    np.linspace(x_min, x_max, 200),
    np.linspace(y_min, y_max, 200)
)
Z = model_pca.predict(np.c_[xx.ravel(), yy.ravel()])
```

```
Z = Z.reshape(xx.shape)

plt.figure()

plt.contourf(xx, yy, Z, alpha=0.3)

plt.scatter(X_pca[:, 0], X_pca[:, 1], c=y)

plt.title("AdaBoost Decision Boundary (PCA Reduced Data)")

plt.xlabel("Principal Component 1")

plt.ylabel("Principal Component 2")

plt.show()
```

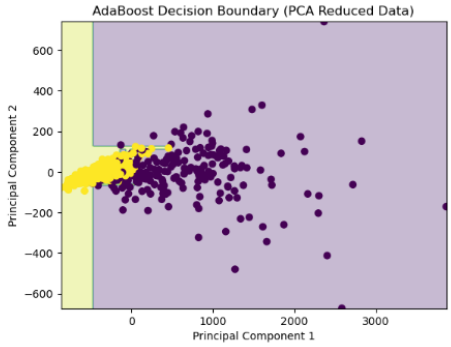
O/P:

```
Accuracy: 0.9736842105263158

Confusion Matrix:
 [[41  2]
 [ 1 70]]

Classification Report:
              precision    recall  f1-score   support

           0       0.98      0.95      0.96        43
           1       0.97      0.99      0.98        71

    accuracy                           0.97       114
   macro avg       0.97      0.97      0.97       114
weighted avg       0.97      0.97      0.97       114

C:\ProgramData\anaconda3\Lib\site-packages\sklearn\ensemble\_weight_boosting.p
  warnings.warn(
```



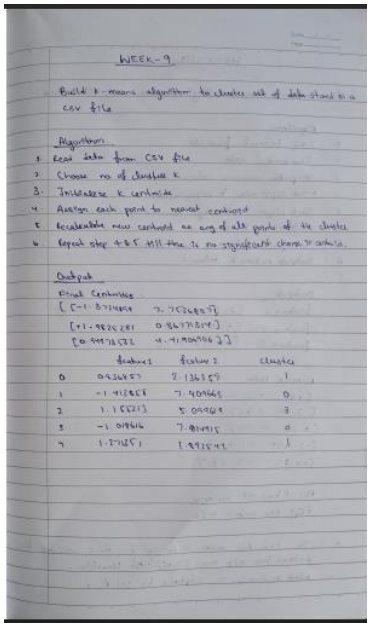



```
C:\ProgramData\anaconda3\Lib\site-packages\sklearn\ensemble\_weight_boosting.py:527: FutureWarning:
  warnings.warn(
```



Program 10

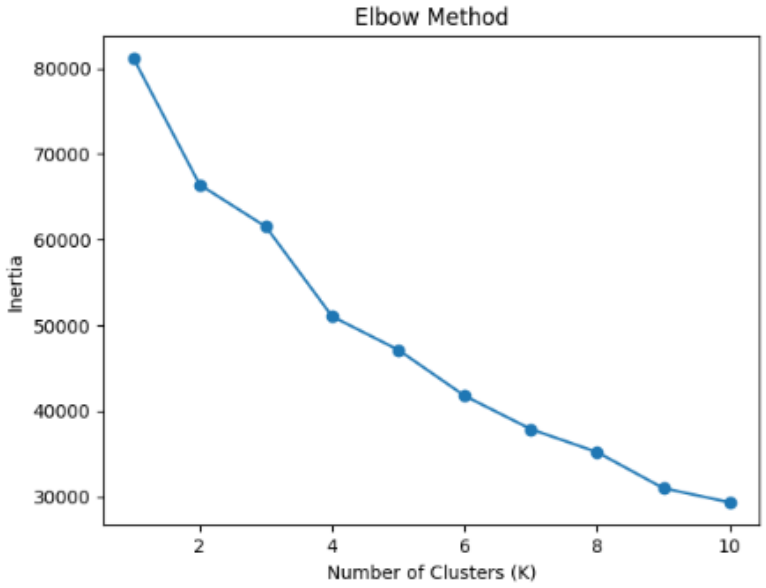
Screenshots



Code:

```
import pandas as pd

from sklearn.preprocessing import StandardScaler, OneHotEncoder

from sklearn.compose import ColumnTransformer

df = X.copy()

# Drop ID

df = df.drop(columns=['status_id'], errors='ignore')

# Convert datetime → extract features

df['status_published'] = pd.to_datetime(df['status_published'])

df['hour'] = df['status_published'].dt.hour

df['day'] = df['status_published'].dt.dayofweek

df = df.drop(columns=['status_published'])
```

```
# Separate categorical & numerical
categorical_cols = ['status_type']
numerical_cols = [col for col in df.columns if col not in categorical_cols]
# ColumnTransformer with OneHotEncoder
preprocessor = ColumnTransformer([
    ('cat', OneHotEncoder(handle_unknown='ignore'), categorical_cols),
    ('num', StandardScaler(), numerical_cols)
])
# Transform data
scaled_data = preprocessor.fit_transform(df)
import matplotlib.pyplot as plt
from sklearn.cluster import KMeans
inertia = []
K_range = range(1, 11)
for k in K_range:
    kmeans = KMeans(n_clusters=k, random_state=42)
    kmeans.fit(scaled_data)
    inertia.append(kmeans.inertia_)
plt.plot(K_range, inertia, marker='o')
plt.xlabel('Number of Clusters (K)')
plt.ylabel('Inertia')
plt.title('Elbow Method')
plt.show()
```

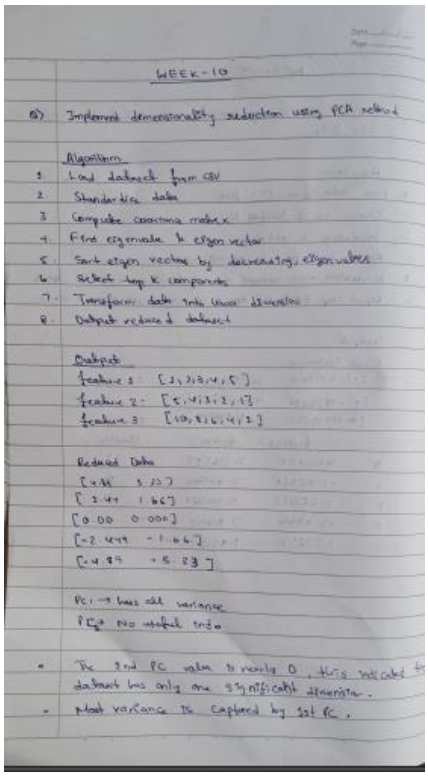


In [27]:

```
from sklearn.cluster import KMeans
from sklearn.decomposition import PCA

# Choose K from elbow
kmeans = KMeans(n_clusters=4, random_state=42)
clusters = kmeans.fit_predict(scaled_data)

# PCA
pca = PCA(n_components=3)
pca_data = pca.fit_transform(scaled_data)

import matplotlib.pyplot as plt

plt.scatter(pca_data[:, 0], pca_data[:, 1], c=clusters)
plt.xlabel('PCA 1')
plt.ylabel('PCA 2')
plt.title('K-Means Clusters')
plt.show()

# Add cluster labels back
df['Cluster'] = clusters
print(df.groupby('Cluster').mean(numeric_only=True))
```

Program 11

Screenshots



WEEK-10

a) Implement dimensionality reduction using PCA method

Algorithm

1. Load dataset from csv
2. Standardize data
3. Compute covariance matrix
4. Find eigenvalue & eigen vector
5. Sort eigen vectors by decreasing eigen values
6. Select top k components
7. Transform data into lower dimensions
8. Output reduced dataset

Outputs

feature 1: [1, 2, 3, 4, 5]
feature 2: [5, 4, 3, 2, 1]
feature 3: [10, 8, 6, 4, 2]

Reduced Data

[4.24 5.12]
[2.44 1.66]
[0.00 0.00]
[-2.449 -1.66]
[-4.24 -5.12]

PC1 → has all variance
PC2 → no useful info

- The 2nd PC value is nearly 0, this indicates the dataset has only one significant dimension.
- Most variance is captured by 1st PC.

Code:

```
import numpy as np
import time
from sklearn.datasets import fetch_california_housing
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
from sklearn.linear_model import LinearRegression
```

```
from sklearn.pipeline import Pipeline
from sklearn.metrics import mean_squared_error
# Load data
data = fetch_california_housing()
X, y = data.data, data.target
# Train-test split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
# -------- WITHOUT PCA --------
start = time.time()
pipe_no_pca = Pipeline([
    ('scaler', StandardScaler()),
    ('model', LinearRegression())
])
pipe_no_pca.fit(X_train, y_train)
y_pred = pipe_no_pca.predict(X_test)
mse_before = mean_squared_error(y_test, y_pred)
time_no_pca = time.time() - start
# -------- WITH PCA (95%) --------
start = time.time()
pipe_pca = Pipeline([
    ('scaler', StandardScaler()),
    ('pca', PCA(n_components=0.95)),
```

```
    ('model', LinearRegression())
])
pipe_pca.fit(X_train, y_train)
y_pred_pca = pipe_pca.predict(X_test)
mse_after = mean_squared_error(y_test, y_pred_pca)
time_pca = time.time() - start
# Extract PCA info
pca_step = pipe_pca.named_steps['pca']
print("===== RESULTS =====")
print(f"Original features: {X.shape[1]}")
print(f"PCA components (95% variance): {pca_step.n_components_}")
print(f"Explained variance: {pca_step.explained_variance_ratio_.sum():.4f}")
print("\n--- Model Performance ---")
print(f"MSE before PCA: {mse_before:.4f}")
print(f"MSE after PCA:  {mse_after:.4f}")
print("\n--- Training Time ---")
print(f"Time without PCA: {time_no_pca:.4f} sec")
print(f"Time with PCA:    {time_pca:.4f} sec")
# -------- MULTIPLE PCA SETTINGS --------
components = [2, 4, 6, 8]
mse_list = []
for n in components:
    pipe = Pipeline([
        ('scaler', StandardScaler()),
```

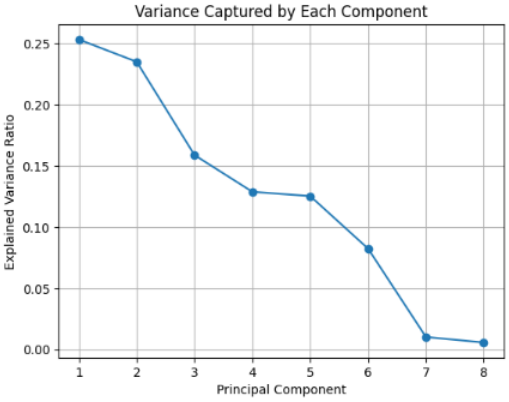
```
        ('pca', PCA(n_components=n)),
        ('model', LinearRegression())
    ])
    pipe.fit(X_train, y_train)
    y_pred_n = pipe.predict(X_test)
    mse_list.append(mean_squared_error(y_test, y_pred_n))
print("\n--- PCA Component vs MSE ---")
for c, m in zip(components, mse_list):
    print(f"Components: {c}, MSE: {m:.4f}")
```

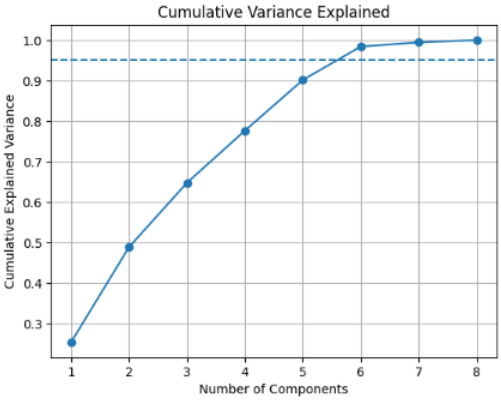
O/P:

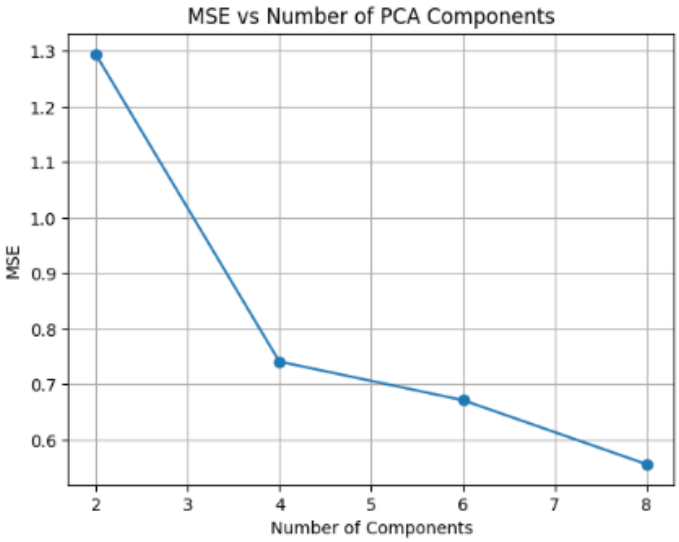
```
Original features: 8
PCA components (95% variance): 6
Explained variance: 0.9837
--- Model Performance ---
MSE before PCA: 0.5559
MSE after PCA:  0.6714
--- Training Time ---
Time without PCA: 0.0501 sec
Time with PCA:    0.0527 sec
--- PCA Component vs MSE ---
Components: 2, MSE: 1.2946
Components: 4, MSE: 0.7409
Components: 6, MSE: 0.6714
Components: 8, MSE: 0.5559
```

```
import matplotlib.pyplot as plt
import numpy as np
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
# Scale full data for visualization
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
# Fit PCA
pca_full = PCA()
pca_full.fit(X_scaled)
explained = pca_full.explained_variance_ratio_
cumulative = np.cumsum(explained)
# ---- Plot 1: Individual variance ----
plt.figure()
plt.plot(range(1, len(explained)+1), explained, marker='o')
plt.xlabel("Principal Component")
plt.ylabel("Explained Variance Ratio")
plt.title("Variance Captured by Each Component")
plt.grid()
plt.show()
# ---- Plot 2: Cumulative variance ----
plt.figure()
```

```
plt.plot(range(1, len(cumulative)+1), cumulative, marker='o')
plt.xlabel("Number of Components")
plt.ylabel("Cumulative Explained Variance")
plt.title("Cumulative Variance Explained")
plt.axhline(y=0.95, linestyle='--')  # 95% threshold
plt.grid()
plt.show()
# ---- Plot 3: PCA components vs MSE ----
plt.figure()
plt.plot(components, mse_list, marker='o')
plt.xlabel("Number of Components")
plt.ylabel("MSE")
plt.title("MSE vs Number of PCA Components")
plt.grid()
plt.show()
```


Variance Captured by Each Component
Explained Variance Ratio
0.25
0.20
0.15
0.10
0.05
0.00
1
2
3
4
5
6
7
8
Principal Component


Cumulative Variance Explained
Cumulative Explained Variance
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
1
2
3
4
5
6
7
8
Number of Components


MSE vs Number of PCA Components
MSE
1.3
1.2
1.1
1.0
0.9
0.8
0.7
0.6
2
3
4
5
6
7
8
Number of Components